18. Off-grid energy — charging Oracle and working without mains power

Oracle OS guide. Manuals in the library: "Electricity for the Farm" (water wheel and dynamo), "Windmills", Energypedia, Appropedia, Practical Action (solar PV, hand pumps).

1. Priority: keep Oracle alive

- **Consumption**: a phone uses ~10–20 Wh per full charge. A 20 W USB solar panel in direct sun gives ~60–80 Wh per day → several charges.
- **USB solar panel**: face the sun (south in the northern hemisphere), tilt ~ your latitude (Scotland ~55°, Portugal ~40°), no shade; re-aim 2–3 times a day. On cloudy days it gives 10–25%.
- **Power bank**: charge the power bank by day and the phone at night from it (protects the phone from fluctuations). LiFePO4 lasts 2,000–4,000 cycles; ordinary lithium 300–500.
- **Batteries**: store at 40–80% charge; avoid heat (never in sun behind glass) and extreme cold while charging (below 0 °C causes damage).
- **Save**: airplane mode, minimum brightness, Oracle's extreme power-saving mode, local AI only when needed (it drains a lot), switch the phone off when not in use.
- **Redundancy**: a spare phone switched off in a metal bag/closed tin (Faraday), charged to 50% and checked every 6 months.

2. Making electricity without a panel

- **Bicycle dynamo**: a hub dynamo (6 V, 3 W) with a dynamo USB charger charges a phone in 2–4 h of pedalling. The bike can be mounted on a fixed stand.
- **Car alternator / DC motor**: turned by crank, water wheel or wind, charges a 12 V battery; then a 12 V→USB converter (car socket).
- **Car batteries**: store energy; don't discharge below ~50% (12.2 V); lead-acid gives off hydrogen — charge outdoors.
- **Charge controller**: always between generator/panel and battery, or it will be destroyed.

3. Mechanical power (what moves a society)

- **Human power**: cranks, pedals, treadle loom and lathe, hand pumps (see Practical Action).
- **Animal traction**: horses, donkeys, oxen — ploughing, hauling, turning mills (horse gin).
- **Water wheel**: needs a drop or steady current. An overshot wheel is most efficient with little water and a 2–10 m drop; undershot for a wide river with little drop. It grinds grain, saws, pumps, powers trip hammers and generates electricity.
- **Windmill**: canvas or wooden sails on a frame facing the prevailing wind; grinding and pumping water.

- **Levers, pulleys, ramps, wheels** — see "Ropes and knots" (mechanical advantage).

4. Heat and light

- **Firewood**: season 1–2 years; green wood gives little heat and lots of smoke. Oak, ash and beech are best.
- **Charcoal**: burns hotter and cleaner (forge, cooking) — see "Basic chemistry".
- **Rocket stove and hay box** (retained-heat cooking) save half the wood — see "Natural building" and "Heat and cook with no power".
- **Light**: tallow or wax candles, vegetable-oil lamps with cotton wicks, dynamo torches; rise and sleep with the sun.
- **Solar hot water**: black hose or a drum painted black in the sun.

5. Biogas (communities, year 2+)

Manure + water in a sealed digester makes methane for cooking; needs careful building (leaks are dangerous). See Energypedia/Appropedia.